

Map to the core portfolios of work covering: Digital Supply Chain, Shop Floor & MES, MRO & Predictive Maintenance, On-Demand Repair, Quality Intelligence (From inspection-based detection to AI-powered prediction and prevention), and Customer Support

The cross-reference matrix below maps key artifacts and governance reviews across the six lifecycle phases (**Discover, Define, Design, Develop, Delivery, Run**) for each core enterprise portfolio in a luxury jet manufacturing environment.

1. Digital Supply Chain Portfolio

Phase	Required Artifacts	Mandatory Reviews & Gates
Discover	Supply Chain Maturity Assessment, Control Tower Business Case	Gate 0: Portfolio Strategy Alignment
Define	Supplier Integration Architecture, EDI/API Data Standards, Tier-1/2 Risk Models	SRR: System Requirements Review
Design	Digital Twin Supply Chain Specs, Vendor Portal UX, Track-and-Trace Data Schemas	PDR / CDR: System Architecture Design Review
Develop	ERP/PLM-Supplier Integration Code, Automated AS9102 FAI Ingestion Workflows	TRR: Supply Chain Integration Test Review
Delivery	Control Tower Go-Live Playbook, Serialized Component Tracking Sign-offs	Deployment Review: Supply Chain Operations Approval
Run	Real-Time Supplier Lead-Time Dashboards, Automated Vendor Risk Scoring Logs	Quarterly Vendor Governance Review

2. Shop Floor & MES (Manufacturing Execution Systems) Portfolio

Phase	Required Artifacts	Mandatory Reviews & Gates
Discover	Industry 4.0 Vision Document, Automation & Robotics Feasibility Study	Gate 0: Shop Floor Modernization Approval

Define	MES Functional Specs, IoT Sensor Integration Specs, Tooling Tracking ConOps	SRR / SFR: MES System Architecture Review
Design	Electronic Work Instructions (eWI) Specs, Line-Balancing Digital Twin Models	PDR / CDR: MES Design Freeze
Develop	MES Core Workflows, PLC/IoT Connectivity Protocols, Smart Tool Calibration Code	TRR: Pilot Line & Integration Readiness Review
Delivery	Factory Acceptance Testing (FAT) Reports, Shop Floor Operator Training Kits	Gate 3 / G3: Assembly Line Go-Live Gate
Run	Real-time OEE (Overall Equipment Effectiveness) Reports, Continuous Line Balance Logs	Monthly Operations & Manufacturing Review

3. MRO & Predictive Maintenance Portfolio

Phase	Required Artifacts	Mandatory Reviews & Gates
Discover	Aircraft Health Management System (AHMS) Market Study, Telemetry Business Case	Gate 0: In-Service Analytics Investment Gate
Define	Telemetry Data Lake ConOps, Fleet Maintenance Requirements, FAA/EASA Data Strategy	SRR: AHMS Functional & Safety Requirements Review
Design	Predictive Failure ML Models, Sensor Telemetry Data Models, DO-178C/DO-254 Specs	PDR / CDR: Predictive Algorithms & Hardware Design
Develop	AHMS Model Validation Reports, MRO Portal Code, Telemetry Pipeline Build	TRR / FTRR: Flight Test Telemetry Validation
Delivery	CAMP (Continued Airworthiness Maintenance Plan) Digital Integration Sign-off	Gate 4: Entry-Into-Service (EIS) Maintenance Readiness

Run	Real-Time Anomaly Alerts, Fleet Health Dashboards, Continuous ML Retraining Logs	CARB: Continued Airworthiness Review Board
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4. On-Demand Repair Portfolio

Phase	Required Artifacts	Mandatory Reviews & Gates
Discover	AOG (Aircraft On Ground) Response Benchmark, Mobile Repair Strategy Document	Gate 0: Rapid Service Investment Gate
Define	Mobile Repair Team (MRT) ConOps, Rapid Engineering Order (REO) Requirements	SRR: On-Demand Service Requirements Review
Design	3D-Printable Spare Part CAD Specs, Rapid Field-Repair Approval Workflows	PDR / CDR: Field Repair Standards Design Review
Develop	Mobile AOG Service App, Automated REO Generation Pipeline, Spares Dispatch Hub	TRR: Field Repair Simulation & App Testing
Delivery	AOG Command Center Playbook, Field Tooling & Logistics Certification Pack	EIS Gate: Service Network Deployment Sign-off
Run	Mean Time to Repair (MTTR) Logs, AOG Dispatch Metrics, Field Repair Quality Audits	Monthly AOG & Field Service Governance

5. Quality Intelligence Portfolio (Inspection $\rightarrow$ AI Prediction & Prevention)

Phase	Required Artifacts	Mandatory Reviews & Gates
Discover	Quality 4.0 Vision, Computer Vision & Acoustic Sensing AI Feasibility Study	Gate 0: Quality Transformation Gate

Define	Digital QMS Architecture, Non-Conformance Taxonomy, Defect Prediction Specs	SRR: Quality Intelligence Requirements Review
Design	AI Defect Detection Models, Automated NDT (Non-Destructive Testing) Sensor Specs	PDR / CDR: Quality Model & Sensor Hardware Design
Develop	Computer Vision Training Datasets, Closed-Loop CAPA Automation Pipelines	TRR: Quality Gate Validation Review
Delivery	Certificate of Conformance (CoC) Automated Generation Sign-offs, Regulatory Audits	Airworthiness Review: FAA/EASA Quality Sign-off
Run	Real-Time Defect Prediction Dashboards, 8D Root Cause Analysis Logs, AI Drift Reports	Quality Board Review: AS9100 Oversight Audit

6. Customer Support Portfolio

Phase	Required Artifacts	Mandatory Reviews & Gates
Discover	VIP Jet Owner Experience Benchmark, Digital Cabin Support Strategy	Gate 0: Customer Experience Investment Approval
Define	Customer Portal Requirements, Custom Cabin Interior Maintenance Specs, Warranty Rules	SRR: Customer Portal & Support Requirements
Design	Jet Owner App Architecture, Interactive Electronic Technical Pubs (IETP) Spec	PDR / CDR: Customer Systems Design Freeze
Develop	VIP Owner Portal, 24/7 Global Support Desk Integration, Warranty Engine Build	TRR: Support Desk Simulation & Portal User Testing
Delivery	Jet Acceptance Digital Checklist, Crew Training Manuals, Warranty Certificates	Final Customer Acceptance Review

Run	Net Promoter Score (NPS) Dashboards, Flight Hour & Warranty Claims Analytics	Annual Owner Operations Review
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